SMART DEVICE TO IMPROVE STRENGTHENING EXERCISE FOR NON-CRITICAL PATIENTS

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INTRODUCTION

BACKGROUND OF THE STUDY

Hand rehabilitation is essential for individuals with physical impairments to restore functional ability in daily activities. The **hand** is one of the most frequently affected body parts due to injuries and medical conditions. **Non-critical patients**, such as individuals with paresis after stroke, sarcopenia, and dynapeniabenefit significantly from progressive resistance training (PRT) to improve strength and functional recovery.



INTRODUCTION

PROBLEM STATEMENT



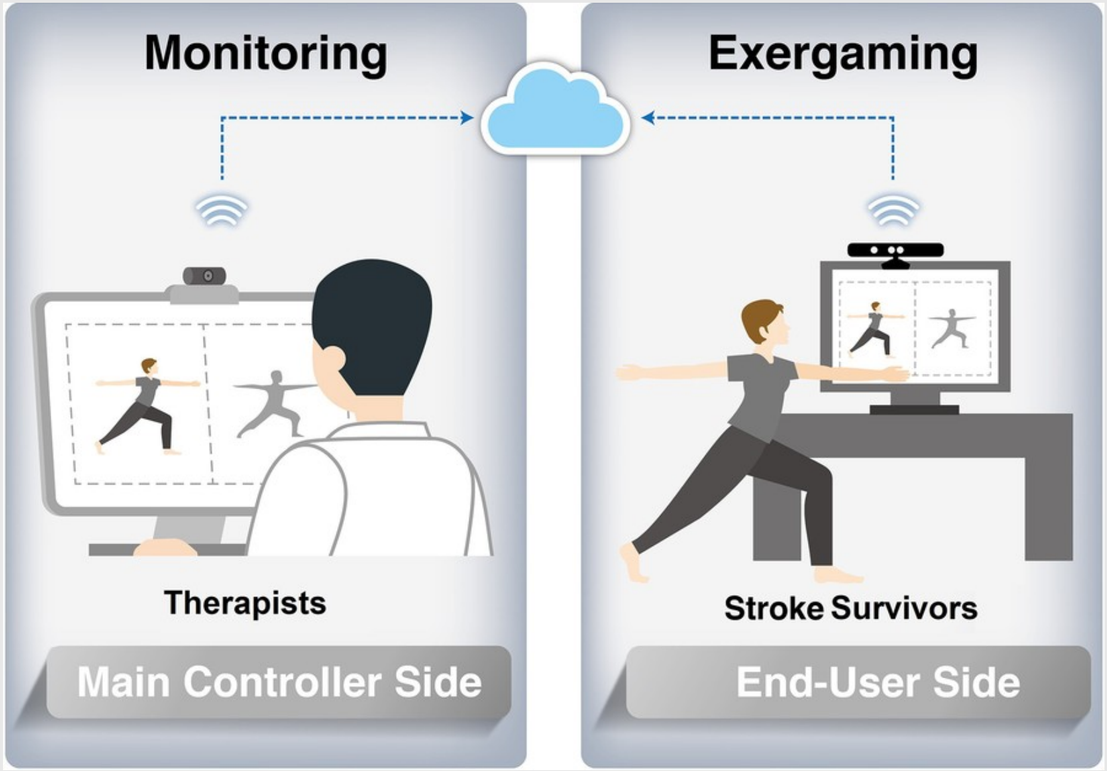


Conventional methods of PRT involve professional who will supervise the patient, uses more than one instrument to evaluate and treat patient.

OVERVIEW

- Telerehabilitation has been a significant solution for a wide range of patients who are not capable of a clinical visit due to cost and accessibility.

- Smart devices are a way to conduct telerehabilitation since they have sensor-integrated equipment, provide real-time feedback, and can track patient progress to help therapists monitor remotely



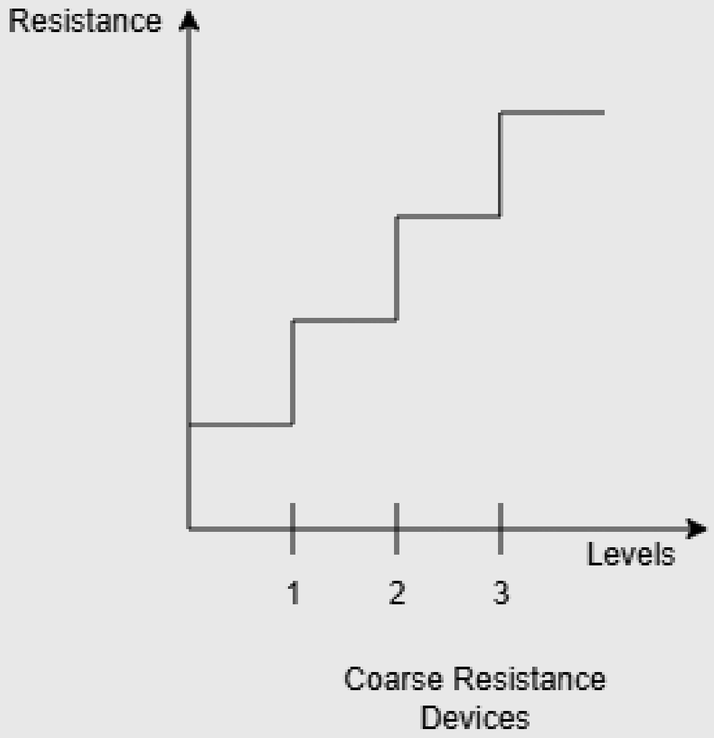
PROPOSED SOLUTION

ELECTROMAGNETISM + TACTILE SENSORS

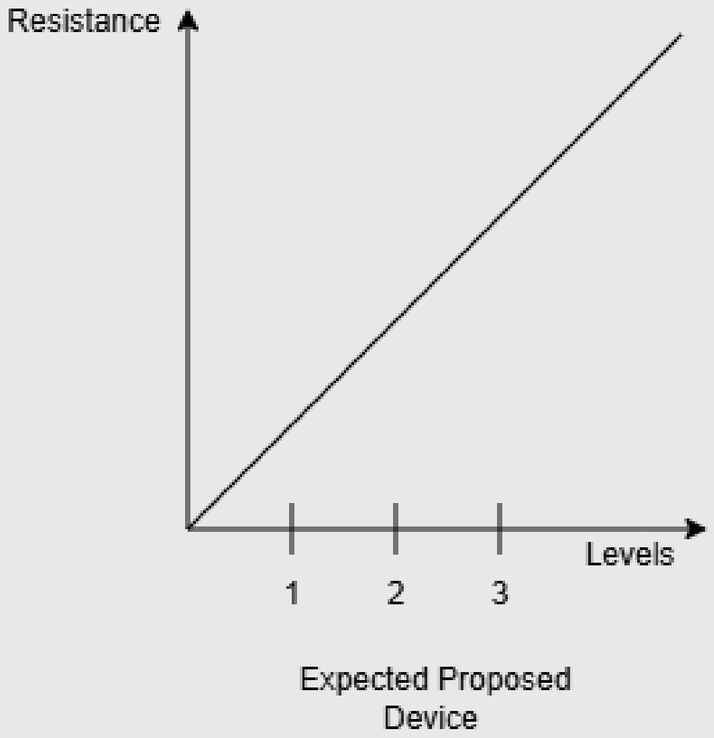
- Commonly used resistance medium of PRTs involve springs and rubber
- Using the concept of electromagnetism we can utilize the repulsion between the two magnets as an alternative resistance medium when encased in silicone rubber
- The silicone rubber provides a protective barrier against the electrical current. In addition, tactile sensors are placed near the opposing magnets which are used to detect grip strength

PROPOSED SOLUTION

ELECTROMAGNETISM + TACTILE SENSORS



Coarse Resistance Devices



Expected Proposed Device

- As illustrated in this figure, the proposed device addresses the fundamental shortcomings of existing coarse-level resistance systems by providing continuous resistance control and intelligent automation, thereby supporting more precise and effective progressive resistance training for non-critical patients.

GOALS AND OBJECTIVE

- Validate the concept of Electromagnetism as a reliable alternative for resistance control as a rehabilitation tool
- Determine the appropriate stiffness of the silicone rubber for a non-critical patient to conduct gripping exercise
- Accurately collect data with the use of tactile sensors to validate the use of electromagnetism and to monitor a patient recovery progress
- Create a Machine Learning algorithm to provide a suitable PRT for the patient using the device
- A web-based application will be developed to display the data collected from the smart rehabilitation device. The system will store and organize the recorded data to enable continuous monitoring, visualization, and assessment of patient rehabilitation progress over time

SCOPE AND LIMITATIONS

- The study focuses on the design and feasibility to create a smart device that serves as a recovery tool that uses Electromagnetism for resistance adjustment intended for hand rehabilitation for a non-critical patient (Paresis, Sarcopenia, Dynapenia). Tactile sensors are used to capture feedback response to collect data and to aid in monitoring the progress.

- This study is limited to the assessment of grip strength in hand rehabilitation, including the analysis and monitoring of the hand, fingers, and wrist. The smart device is intended to precisely measure grip force and grip repetitions of non-critical patients performing hand-gripping exercises. A web-based application is utilized for data collection, visualization, and analysis of the rehabilitation data.

SIGNIFICANCE OF THE STUDY

THIS STUDY AIMS TO DEVELOP AND TEST A PROTOTYPE SMART GRIP RECOVERY TOOL THAT ADDRESSES ACCESSIBILITY BARRIERS IN HAND REHABILITATION BY ENABLING SAFE, ADJUSTABLE RESISTANCE TRAINING AT HOME OR IN CLINICS.

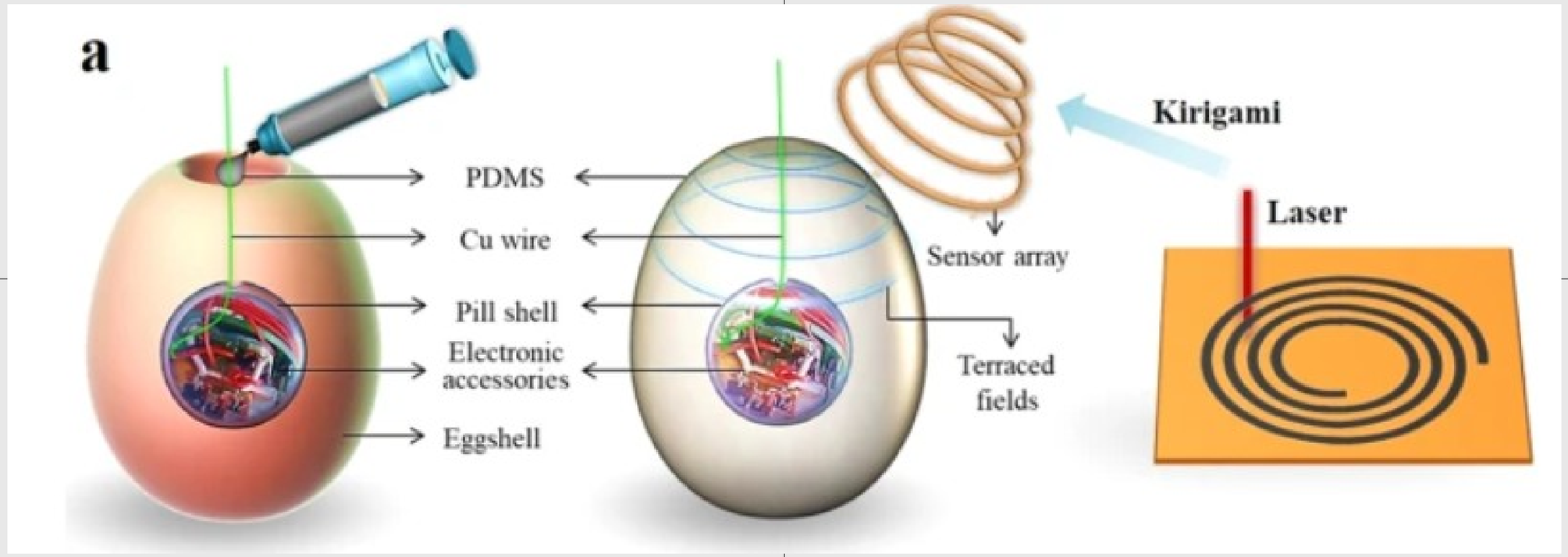
- Elderly/mobility-impaired patients (Sarcopenia, Dynapenia)
- Post-injury rehabilitation users needing grip strength tracking
- Stroke survivors with Paresis
- Therapists using remote data monitoring

REVIEW OF RELATED LITERATURE

The introduction of using smart devices for rehabilitation has significantly transformed the rehabilitation process for strengthening training for non-critical patients by enabling them to conduct exercises by objective monitoring

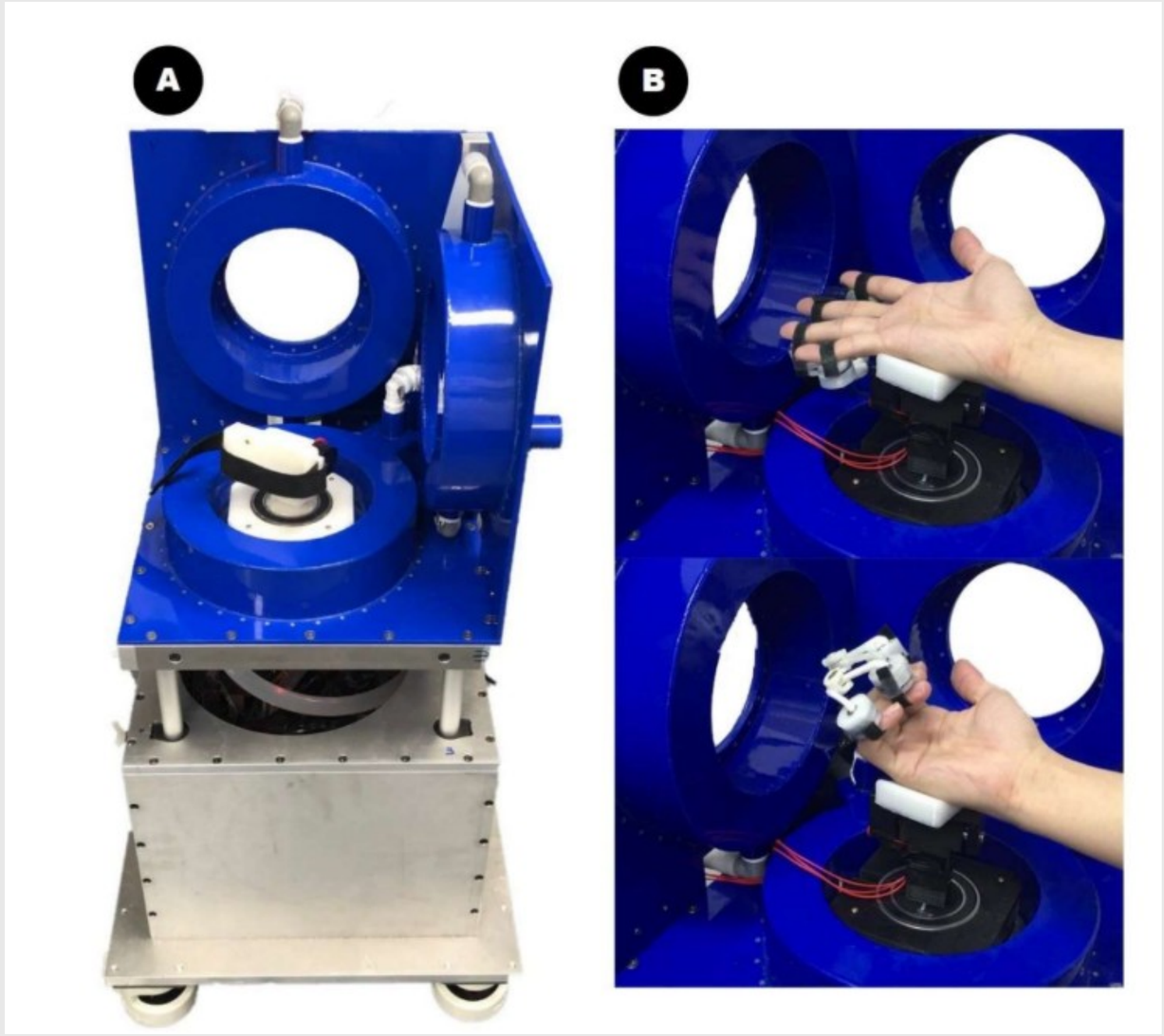
A SMART-GRIP STRENGTH MONITORING DEVICE IN THE FORM OF A SENSOR-INTEGRATED BALL USING LASER-KIRIGAMI GRAPHENE TECHNOLOGY

THREE-DIMENSIONAL MAGNETIC FINGER REHABILITATION DEVICE AIMED AT ASSISTING FINGER MOVEMENT THROUGH CONTROLLED ELECTROMAGNETIC FORCE



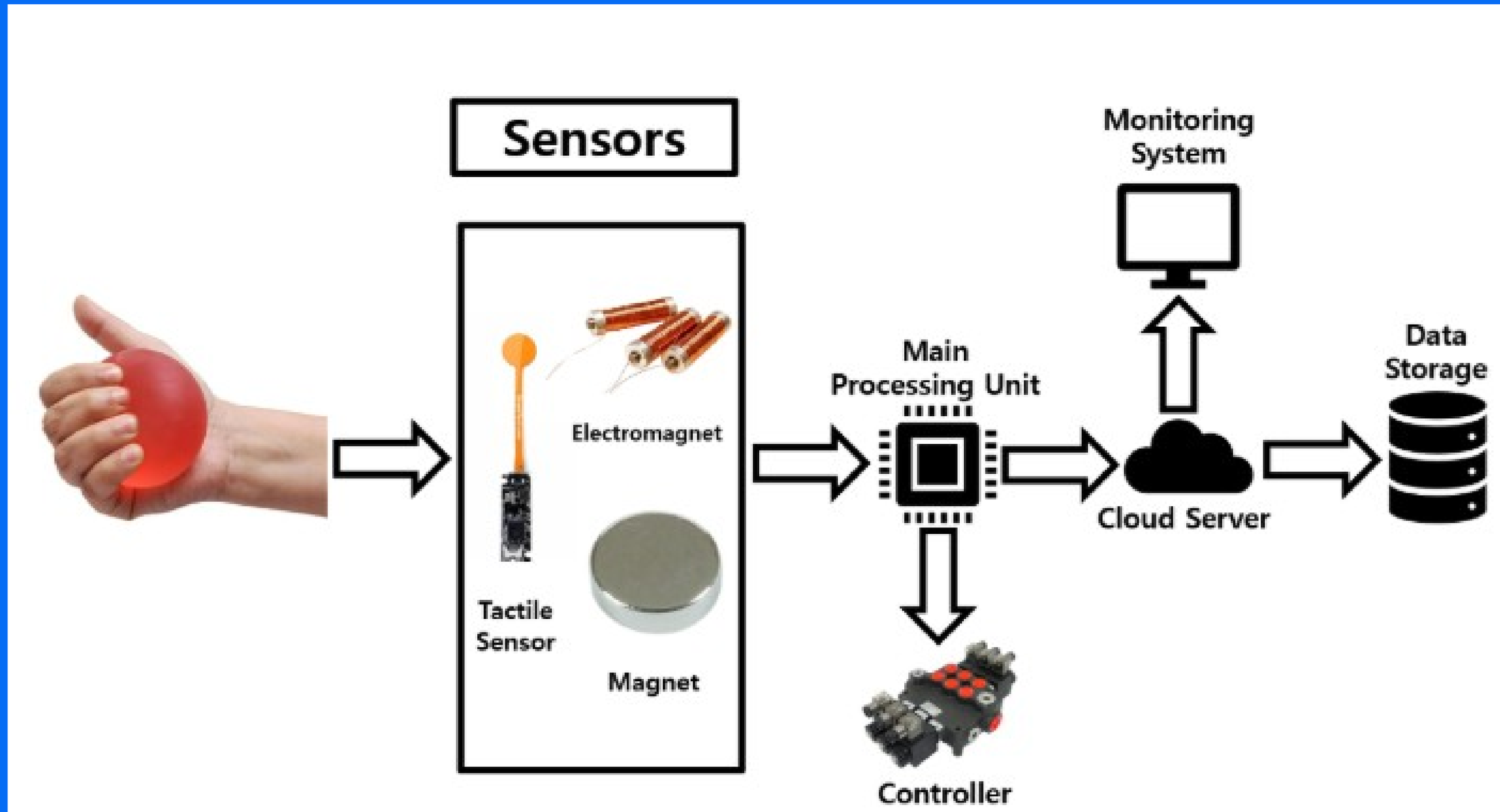
A smart ball sensor fabricated by laser kirigami of graphene for personalized long-term grip strength monitoring

Three-Dimensional Magnetic Rehabilitation, Robot-Enhanced Hand-Motor Recovery after Subacute Stroke: A Randomized Controlled Trial

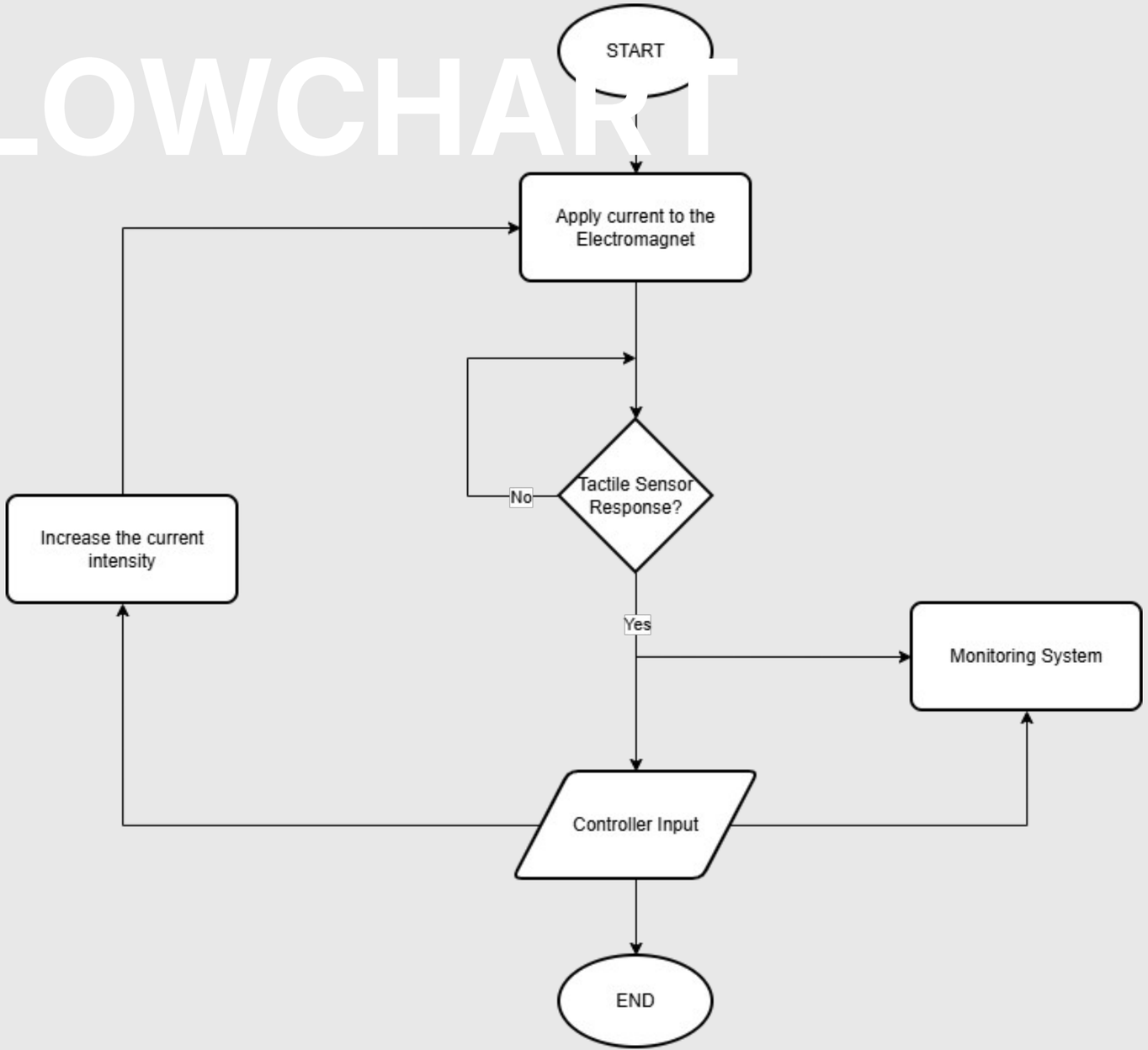


METHODOLOGY

CONCEPTUAL FRAMEWORK

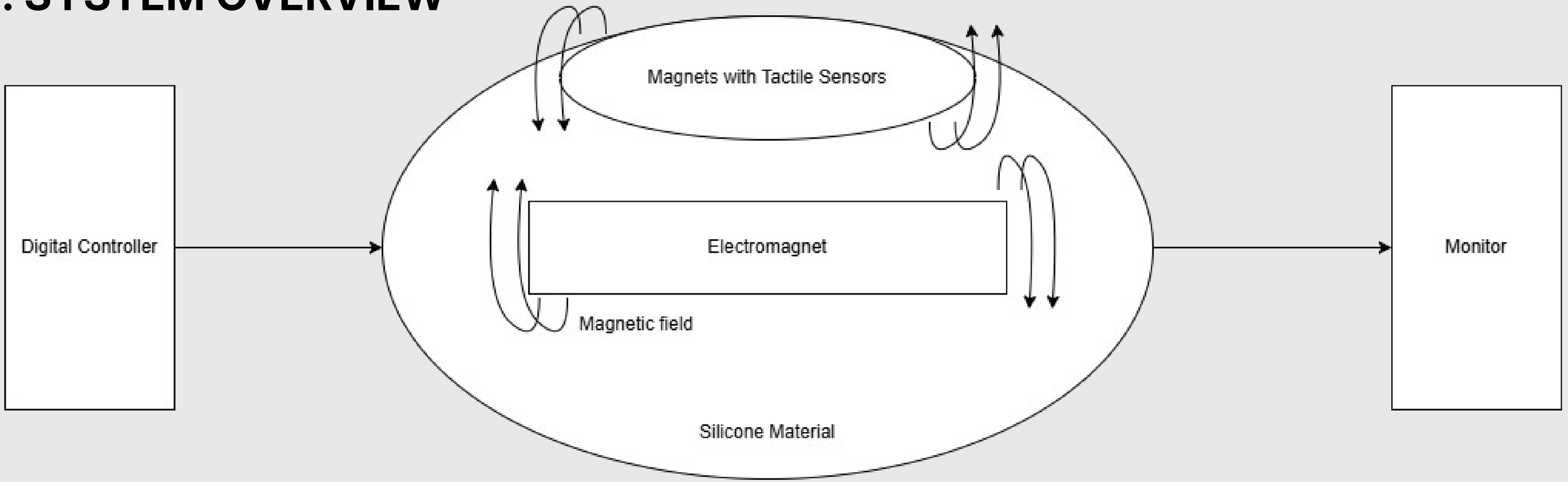


SYSTEM FLOWCHART



SYSTEM ANALYSIS AND DESIGN

1. SYSTEM OVERVIEW



SYSTEM ANALYSIS AND DESIGN

2. SILICONE RUBBER EXPERIMENTATION

- The initial phase of system development involves experimental evaluation to determine the appropriate Shore A hardness of silicone rubber with respect to flexibility, malleability, and heat resistance.
- The objective of this phase is to identify an optimal Shore A value that provides the lowest possible stiffness suitable for non-critical patients, while still maintaining adequate flexibility, malleability, and heat resistance.

SYSTEM ANALYSIS AND DESIGN

3. ELECTROMAGNETISM IMPLEMENTATION

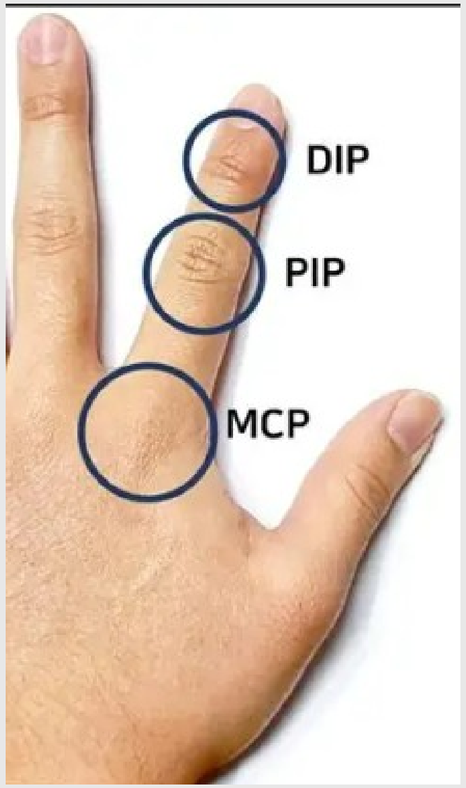
- This phase focuses on identifying the most suitable electromagnet for integration into the system, including an evaluation of solenoid magnets, toroidal magnets, and other applicable electromagnet configurations
- Baseline measurements are first obtained with zero applied current. Subsequently, the current is incrementally increased while maintaining a constant applied pressure, and the resulting changes in system height are recorded.

SYSTEM ANALYSIS AND DESIGN

4.TACTILE SENSOR IMPLEMENTATION

- Tactile sensors are embedded beneath finger-groove surfaces to ensure consistent finger contact and repeatable pressure measurements. The silicone layer distributes load, reduces discomfort, and reliably transmits deformation to the sensors during squeeze and release.

- Sensor signals are continuously sampled and time-stamped, with filtering applied to reduce noise and baseline drift. Key features such as peak pressure and cycle-to-cycle variability are extracted, while full encapsulation and strain relief ensure long-term stability and durability



SYSTEM ANALYSIS AND DESIGN

5.INTEGRATION OF ALL SYSTEMS

- The electromagnet module generates adjustable resistance through controlled current levels, while the tactile sensors simultaneously capture grip response. Resistance settings and pressure signals are time-stamped to create paired load–response data for each repetition.

- Electrical noise from electromagnet switching is minimized using grounding, shielding, and signal filtering. Processed tactile features and raw sensor data are transmitted to a monitoring system for real-time visualization, session logging, and post-exercise performance review.

SYSTEM DEVELOPMENT, TESTING AND DEPLOYMENT STRATEGIES

- Target: Healthy males 20–29
- Exclusion: Recent hand/finger injury/surgery, hand pain, musculoskeletal or neurological issues
- Task: Squeeze–release trials with smart grip at 3 resistance levels
- Measure: Tactile signals + resistance, usability questionnaire
- Success: Stable tactile data, consistent response to resistance, easy task completion

ANNEX

WORKPLAN

WORKPLAN AND EXPECTED OUTPUT

(NOTE: Include activity/tasks till end of 2nd sem or end of project/research. Append Gantt chart when appropriate)

Activity/Task	***Wk 1-4***	***Wk 5-8***	***Wk 9-12***	***Wk 13-16***	***Wk 17-44***	***Expected Output***	***Person Assigned***
[1] Literature Review & Conceptual Refinement	XXXX					Updated Conceptual Framework	USC & SCH
[2] Technical Design & System Architecture		XXXX				Final System Flowchart	USC & SCH
[3] Drafting Methodology & Experimental Design			XXXX			Comprehensive Research Paper Draft	USC & SCH
[4] Paper Finalization & Internal Review				XXXX		Finalized Technical Paper	USC & SCH
[5] Silicone Rubber Experimentation					X 17-20	Optimal Shore A material selection	USC
[6] Electromagnetism System Implementation					X 21-24	Tested magnetic actuator system	USC
[7] Tactile Sensor Integration & Encapsulation					X 25-28	Assembled tactile sensing layer	SCH
[8] Integration of All Systems					X 29-32	Functional "Smart Grip" Prototype	USC & SCH
[9] Cloud Server & Monitoring System Setup					X 33-36	Real-time data logging & visualization interface	SCH
[10] System Validation & Participant Testing					X 37-40	Recorded tactile signals & resistance logs	USC & SCH
[11] Data Analysis & Project Completion					X 41-44	Final Project Report & Usability Results	USC & SCH

ANNEX

PROJECT COST ESTIMATES

COST ESTIMATES

(NOTE: In making the cost estimates please include contingency which is about 20% of the total cost.)

Proposed Budget for Materials

Item Description	*Quantity*	*Cost/Unit*	*Subtotal*
[1] Tactile Sensors (FSR/Capacitive Array)	**4 units**	**1,200.00**	**4,800.00**
[2] High-Torque Electromagnets	**1 unit**	**2,800.00**	**2,800.00**
[3] Neodymium Permanent Magnets	**2 units**	**350.00**	**700.00**
[4] Main Processing Unit (ESP32-S3)	**1 unit**	**450.00**	**450.00**
[5] Silicone Rubber Kit (Shore A grade)	**2 kg**	**1,100.00**	**2,200.00**
		Total	**10,950.00**

Proposed Budget for Laboratory Equipment/Apparatus Use

Item Description	*Quantity*	*Cost/Unit*	*Subtotal*
[1] Hydraulic Press Machine (Rental/Session)	**5 sessions**	**2,000.00**	**10,000.00**
[2] Ultrasonic Sensor (Displacement Testing)	**1 unit**	**250.00**	**250.00**
		Total	**10,250.00**

Miscellaneous Costs

Items			
Description	*Quantity*	*Cost/Unit*	*Subtotal*
[1] External Material Analysis (Stress Test)	**1 lot**	**8,500.00**	**8,500.00**
[2] Cloud Server & Data Storage (1 Year)	**1 lot**	**6,000.00**	**6,000.00**
[3] Participant Honorariums (20 subjects)	**20 pax**	**500.00**	**10,000.00**
[4] Consent Forms & Documentation Printing	**1 lot**	**1,500.00**	**1,500.00**
		SUBTOTAL	**26,000.00**

ANNEX

BUGET SUMMARY

Budget Summary

Description	Amount
Total Project Cost	**47,200.00**
Contingency Fee (20%)	**9,440.00**
GRAND TOTAL	**56,640.00**

THANK YOU

OPEN FOR DISCUSSION